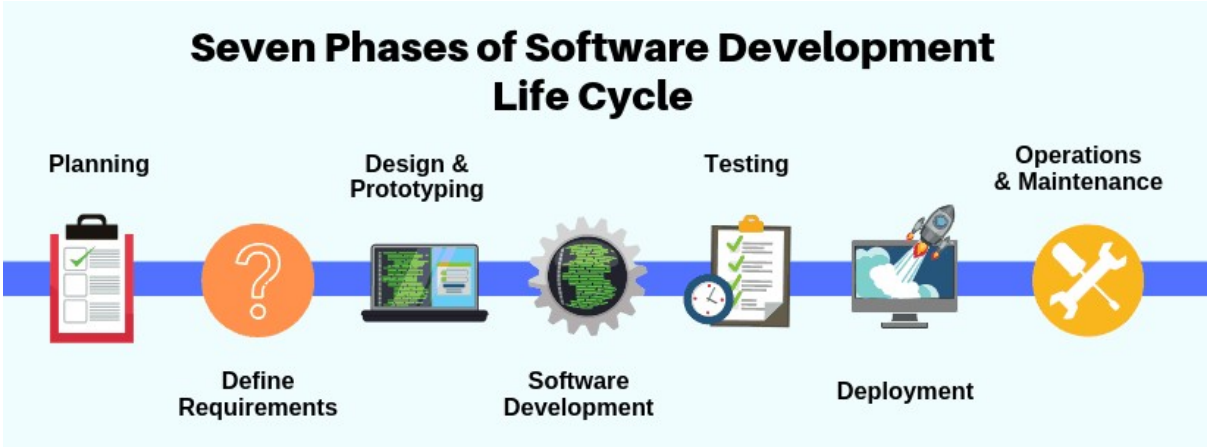
System development using the concept of SDLC:

SDLC, or Software Development Life Cycle, is a set of steps used to create software applications. These steps divide the development process into tasks that can then be assigned, completed, and measured.

It's typically divided into six to eight steps:

- Planning
- Defining Requirements
- Design and Prototyping
- Software Development
- Software Testing
- Deployment
- Operations and Maintenance



1. **Planning**

In the Planning phase, project leaders evaluate the terms of the project.

Planning defines the scope and purpose of the application. It provides a plan for the team to effectively create the software. It sets boundaries to help keep the project from expanding or shifting from its original purpose. This includes calculating and creating a timetable with target goals.

Most importantly, the planning stage sets the project schedule, which can be of key importance if development is for a commercial product that must be sent to market by a certain time.

2. **Define Requirements**

Defining requirements is considered part of planning to determine what the application is supposed to do and its requirements. It:

- Define any prototype system requirements
- Perform research and analysis to determine the needs of end-users

This is the requirements table of our project

ID	Requirements	Needed by Programmer	Solution
101	Log in with Safety Pin	Need to log in user with correct safety pin	Declare void function (login) and put do…while loop to check safety pin Use cin command to input user safety pin
102	Menu system	Need to display all the actions available to the user	Declare void function(menuSystem) and use cout command to display actions
103	Action system	Need to loop the program until the user inputs 0	Use do…while loop and switch loop to repeat the action until user inputs 0
104	Balance Inquiry	Need to check account balance and account number	Use cout command to display user account balance
105	Withdraw from account	Need to withdraw money from own account	Use if…else statement to check if there is enough money in account
106	Transfer from account	Need to transfer money from account to another account	Use if…else statement to check if there is enough money in account Use cin command to input transferred account number
107	Deposit to account	Need to deposit money to own account	Use cin command to deposit money to own account
108	Clear screen	Need to clear screen and display menu	Include <windows.h> Use system(“cls”) to clear the screen and display menu
109	Reload TNG card	Need to reload Touch N’ Go card from account to TNG card number	Use if…else statement to check if there is enough money in account Use cin command to input transferred TNG card number
110	Exit program	Need to end program with thank you message	Use system(“pause”) to pause program before exiting program
111	Char filter	Need to filter out character input from user	Use while loop to filter out characters

3. Design and Prototyping

The Design phase models the way a software application will work. Some aspects of the design include:

User Interface – Defines the ways customers interact with the software, and how the software responds to input.

Programming – Not just the programming language, but including methods of solving problems and performing tasks in the application.

Platforms – Defines the platforms on which the software will run, such as Apple, Android, Windows, Linux.

A prototype is like one of the early versions of software in the Iterative software development model. It demonstrates a basic idea of how the application looks and works.

4. Software Development

The coding process includes many other tasks. Finding and fixing errors and glitches is critical. Tasks often hold up the development process, such as compiling code so an application can run. SDLC can anticipate these delays so that developers can be tasked with other duties.

Software developers appreciate instructions and explanations

5. Testing

It's critical to test an application before making it available to users. Testing should ensure that each function works correctly. The testing phase helps reduce the number of bugs and glitches that users encounter. This leads to a higher user satisfaction and a better usage rate.

6. Deployment

In the deployment phase, the application is made available to users.

7. Operations and Maintenance

At this point, the development cycle is almost finished. The application is done and being used in the field. The Operation and Maintenance phase is still important, though. In this phase, users discover bugs that weren't found during testing. These errors need to be resolved, which can spawn new development cycles.

SDLC Model:

Iterative Model

The iterative model focuses on repetition and repeat testing. New versions of a software project are produced at the end of each phase to catch potential errors and allow developers to constantly improve the end product.

Benefits of SDLC

Clear Goal Descriptions

Developers clearly know the goals they need to meet and the deliverables they must achieve by a set timeline, lowering the risk of time and resources being wasted.

Proper Testing Before Installation

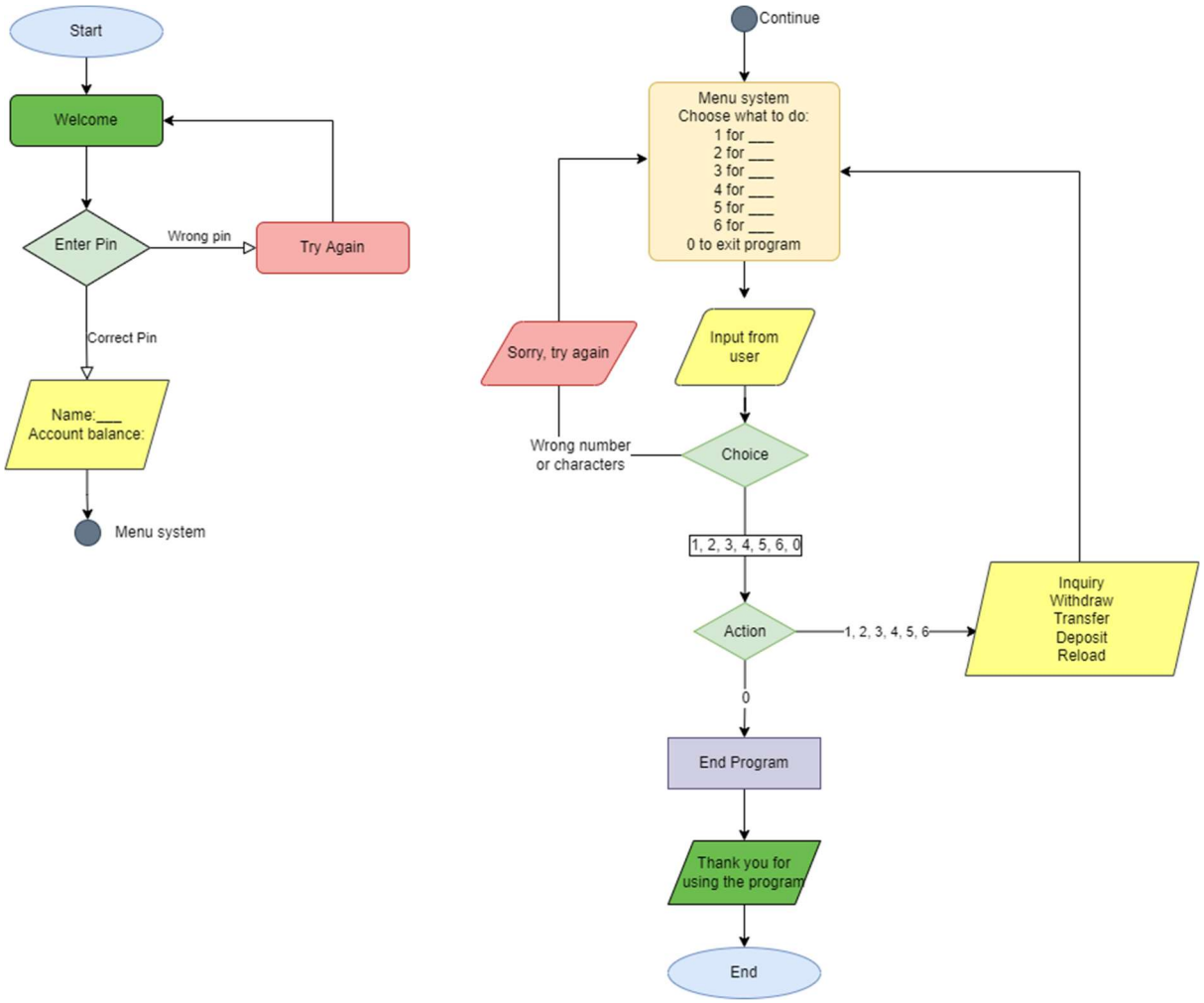
SDLC models implement checks and balances to ensure that all software is tested before being installed in greater source code.

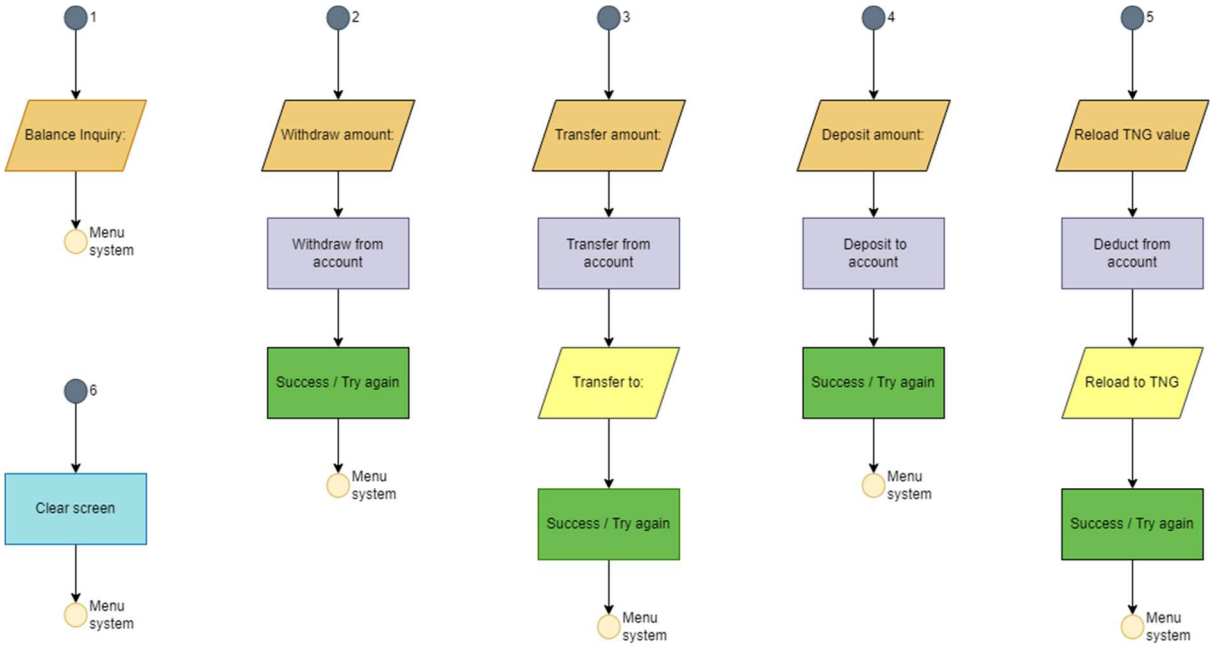
Where is SDLC Used?

System development life cycles are typically used when developing IT projects.

Software development managers will utilize SDLCs to outline various development stages, make sure everyone completes stages on time and in the correct order, and that the project is delivered as promptly and as bug-free as possible.

Flowchart


Start
Welcome
Enter Pin
Wrong pin
Try Again
Correct Pin
Name:___
Account balance:
Menu system
Continue
Menu system
Choose what to do:
1 for ___
2 for ___
3 for ___
4 for ___
5 for ___
6 for ___
0 to exit program
Sorry, try again
Input from user
Wrong number or characters
Choice
1, 2, 3, 4, 5, 6, 0
Action
1, 2, 3, 4, 5, 6
Inquiry
Withdraw
Transfer
Deposit
Reload
0
End Program
Thank you for using the program
End


1
Balance Inquiry:
Menu system
2
Withdraw amount:
Withdraw from account
Success / Try again
Menu system
3
Transfer amount:
Transfer from account
Transfer to:
Success / Try again
Menu system
4
Deposit amount:
Deposit to account
Success / Try again
Menu system
5
Reload TNG value
Deduct from account
Reload to TNG
Success / Try again
Menu system
6
Clear screen
Menu system

Program code:

atm.cpp

```cpp
#include <iostream>
#include <string>
#include <windows.h>
using namespace std;

    //declare all the variables here
    int safety_pin, option, recepient, acc_num;
    float acc_balance, withdraw, transfer, deposit, reload;
    string name;

    //for the users;
    int safety_pin1 = 1971;
    string name1 = "Yi Long Mask";
    int acc_num1 = 2203837;
    float acc_balance1 = 20000.00;

    int safety_pin2 = 6969;
    string name2 = "Zhong Xina";
    int acc_num2 = 80085;
    float acc_balance2 = 420420.00;

    int safety_pin3 = 111;
    string name3 = "Bing Chilling";
    int acc_num3 = 444;
    float acc_balance3 = 111111.00;
```

```cpp
void login()            //login menu
{
    cout<<"Please enter your safety pin: ";
    while(!(cin >> safety_pin))         //filter out chars
    {
        cout <<"This is not integers! Try again!\n";
        cout<<"Enter pin: ";
        cin.clear();
        cin.ignore(123,'\n');
    }

    if (safety_pin != safety_pin1 and safety_pin != safety_pin2 and safety_pin != safety_pin3)
    {
        do
        {
            system("color 0c");     //change the terminal color to red, because it looks cool
            cout<<"\a";
            cout<<"Wrong pin. Try again!\n";
            cout<<"Enter pin: ";
            while(!(cin >> safety_pin))         //filter out chars further, trust me, there was a bug here
            {
                cout <<"This is not integers! Try again!\n";
                cout<<"Enter pin: ";
                cin.clear();
                cin.ignore(123,'\n');
            }
        }while(safety_pin != safety_pin1 and safety_pin != safety_pin2 and safety_pin != safety_pin3);
    }
}
```

```cpp
void showMenu()         //main menu
{
    cout<<"\n-~-~-~-~-~-~-~-~-~-~-~-~-~-\n"
        <<"Choose what to do:\n"
        <<"  [1] - Balance Inquiry\n"
        <<"  [2] - Money Withdrawal\n"
        <<"  [3] - Credit Transfer\n"
        <<"  [4] - Money Deposit\n"
        <<"  [5] - Touch N Go reload\n"
        <<"  [6] - Clear screen\n"
        <<"  [0] - Quit Program\n"
        <<"-~-~-~-~-~-~-~-~-~-~-~-~-~-\n";;
}
```

```
71  void action()           //the looping action of atm
72  {
73
74      cout.setf(ios::fixed);
75      cout.setf(ios::showpoint);
76      cout.precision(2);
77
78      do
79      {
80          showMenu();
81          while(!(cin >> option))     //filter out chars
82          {
83              cout <<"This is not integers! Try again!\n";
84              cout<<"Choose what to do:\n";
85              cin.clear();
86              cin.ignore(123,'\n');
87          }
88
89          switch(option)
90          {
91              case 1:     //account inquiry
92                  system("cls");
93                  cout<<"Balance Inquiry:\n"
94                      <<"Your name: "<<name
95                      <<"\nAccount number: "<<acc_num
96                      <<"\nAccount balance: RM "<<acc_balance<<endl;
97              break;
98
99              case 2:     //withdraw money
100                 cout<<"Withdraw:\n"
101                     <<"Account balance: RM "<<acc_balance
102                     <<"\nWithdraw amount: RM ";
103
104                 while(!(cin >> withdraw))           //filter out chars
105                 {
106                     cout <<"This is not integers! Try again!\n";
107                     cin.clear();
108                     cin.ignore(123,'\n');
109                     cout << "\nWithdraw amount: RM ";
110                 }
111
112                 if (withdraw <= acc_balance)
113                 {
114                     acc_balance = acc_balance - withdraw;
115                         cout<<"\nMoney withdrawal success. The money has been deducted from your account."
116                             <<"\nYour current account balance is: RM "<<acc_balance<<endl;
117                         if (acc_balance == 0)
118                         {
119                             cout<<"You have no money now!!";
120                         }
121                 }
122                 else{
123                     cout<<"The withdrawal amount is too much! Not enough in account balance!";
124                 }
125             break;
```

```
127             case 3:     //transfer credit
128                 cout<<"Transfer:\n"
129                     <<"Account balance: RM "<<acc_balance
130                     <<"\nTransfer amount: RM ";
131 
132                 while(!(cin >> transfer))           //filter out chars
133                 {
134                     cout <<"This is not integers! Try again!\n";
135                     cin.clear();
136                     cin.ignore(123,'\n');
137                     cout << "\nTransfer amount: RM ";
138                 }
139 
140                 if (transfer <= acc_balance)
141                 {
142                     acc_balance = acc_balance - transfer;
143                     cout<<"Transfer to (Enter recipient account number):";
144                     cin>>recepient;
145                     cout<<"\nMoney transfer success. The money has been deducted from your account."
146                         <<"\nYour current account balance is: RM "<<acc_balance<<endl;
147                         if (acc_balance == 0)
148                         {
149                             cout<<"You have no money now!!";
150                         }
151                 }
152 
153                 else{
154                     cout<<"The transfer amount is too much! Not enough in account balance!";
155                 }
156             break;
157 
158             case 4:     //deposit
159                 cout<<"Deposit:\n"
160                     <<"Account balance: RM "<<acc_balance
161                     <<"\nDeposit amount: RM ";
162 
163                 while(!(cin >> deposit))            //filter out chars
164                 {
165                     cout <<"This is not integers! Try again!\n";
166                     cin.clear();
167                     cin.ignore(123,'\n');
168                     cout << "\nDeposit amount: RM ";
169                 }
170 
171                 acc_balance = acc_balance + deposit;
172                 cout<<"\nMoney deposit success. The money has been deposited to your account."
173                     <<"\nYour current account balance is: RM "<<acc_balance<<endl;
174             break;
```

```
            case 5:     //reload TnG card
                cout<<"Reload Touch 'N Go card:\n"
                    <<"Account balance: RM "<<acc_balance
                    <<"\nReload Value: RM ";

                while(!(cin >> reload))         //filter out chars
                {
                    cout <<"This is not integers! Try again!\n";
                    cin.clear();
                    cin.ignore(123,'\n');
                    cout << "\nReload Value: RM ";
                }

                if (reload <= acc_balance){
                    acc_balance = acc_balance - reload;
                    cout<<"Reload to (Enter Touch 'N Go card number):";
                    cin>>recepient;
                    cout<<"\nCard reload success. The money has been deducted from your account."
                        <<"\nYour current account balance is: RM "<<acc_balance<<endl;
                        if (acc_balance == 0){
                            cout<<"You have no money now!!";
                        }
                }

                else{
                    cout<<"The transfer amount is too much! Not enough in account balance!";
                }
            break;

            case 6:     //clear screen
                system("cls");
            break;

            case 0:
                cout<<"Thank you for using this program!"<<endl;
                system("pause");
            break;

            default:
                cout<<"Sorry, this action cannot be done. Please try again.";
            break;
        }
    }while(option != 0);
}
```

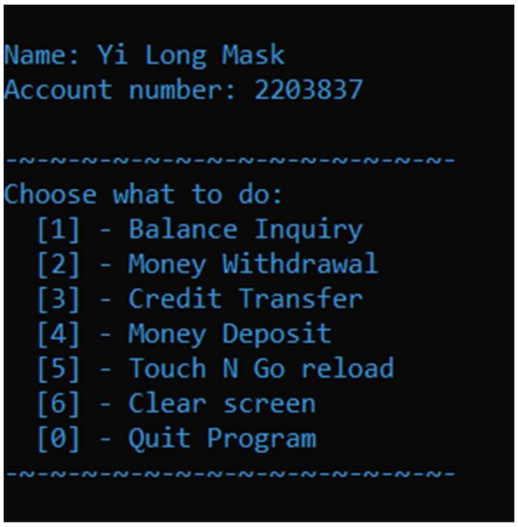
```cpp
int main(){
    system("color 0e");     //change the color of the terminal to yellow
    cout<<"~~~
```

OUTPUT:

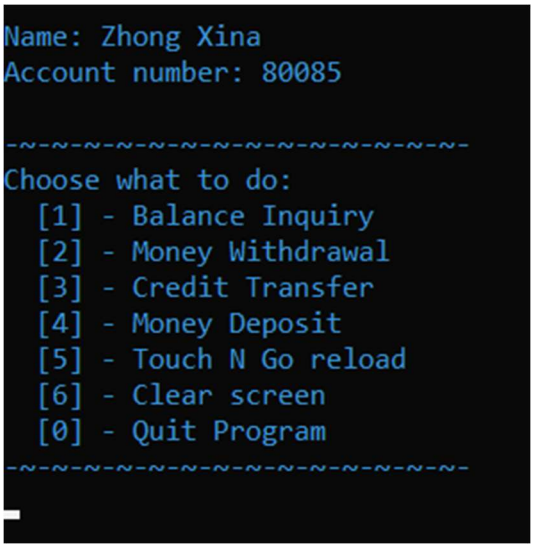
Welcome screen



User 1



User 2



User 3

```
Name: Bing Chilling
Account number: 444

-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-
Choose what to do:
  [1] - Balance Inquiry
  [2] - Money Withdrawal
  [3] - Credit Transfer
  [4] - Money Deposit
  [5] - Touch N Go reload
  [6] - Clear screen
  [0] - Quit Program
-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-
```

Menu system

```
Choose what to do:
  [1] - Balance Inquiry
  [2] - Money Withdrawal
  [3] - Credit Transfer
  [4] - Money Deposit
  [5] - Touch N Go reload
  [6] - Clear screen
  [0] - Quit Program
-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-
```

Option 1 (Balance Inquiry)

```
Balance Inquiry:
Your name: Bing Chilling
Account number: 444
Account balance: RM 111111.00

-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-
Choose what to do:
  [1] - Balance Inquiry
  [2] - Money Withdrawal
  [3] - Credit Transfer
  [4] - Money Deposit
  [5] - Touch N Go reload
  [6] - Clear screen
  [0] - Quit Program
-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-~-
```

Option 2 (Withdraw)

```
2
Withdraw:
Account balance: RM 111111.00
Withdraw amount: RM 3453

Money withdrawal success. The money has been deducted from your account.
Your current account balance is: RM 107658.00
```

Option 3 (Transfer)

```
3
Transfer:
Account balance: RM 107658.00
Transfer amount: RM 2343
Transfer to (Enter recipient account number):2342

Money transfer success. The money has been deducted from your account.
Your current account balance is: RM 105315.00
```

Option 4 (Deposit)

```
4
Deposit:
Account balance: RM 105315.00
Deposit amount: RM 243242

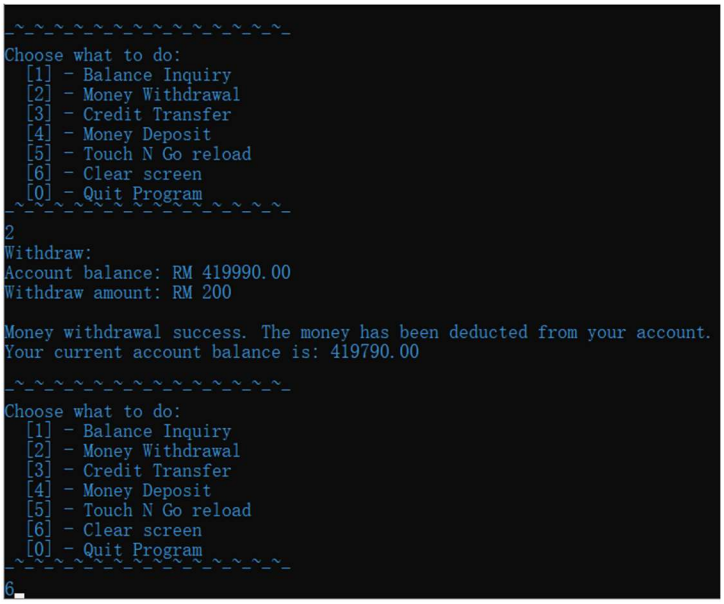
Money deposit success. The money has been deposited to your account.
Your current account balance is: RM 348557.00
```

Option 5 (Reload Touch ‘N Go Card)

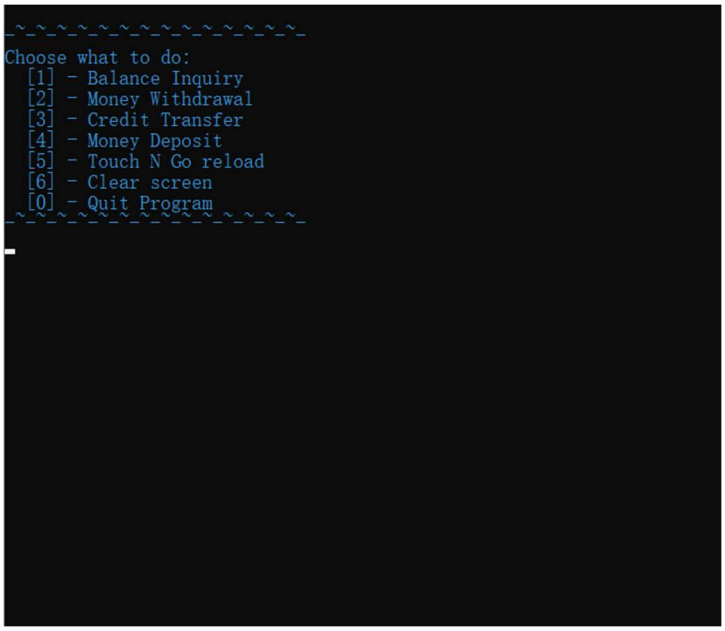
```
5
Reload Touch 'N Go card:
Account balance: RM 348557.00
Reload Value: RM 34534
Reload to (Enter Touch 'N Go card number):796855

Card reload success. The money has been deducted from your account.
Your current account balance is: RM 314023.00
```

Option 6 (Clear Screen)

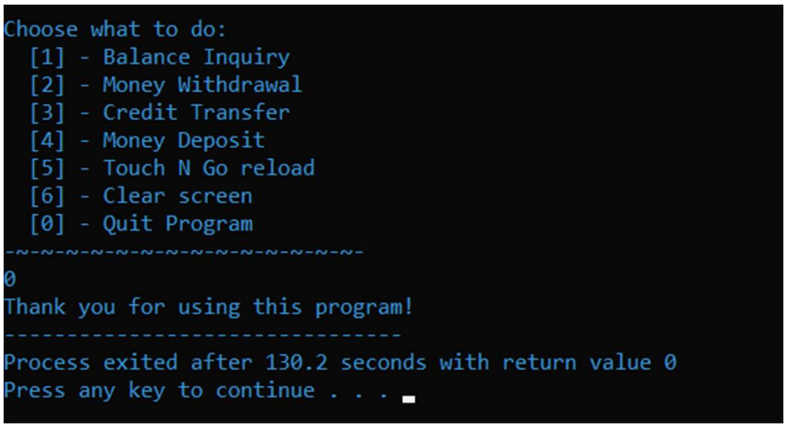


(Before)



(After)

Option 0: Thank You screen (exit program)

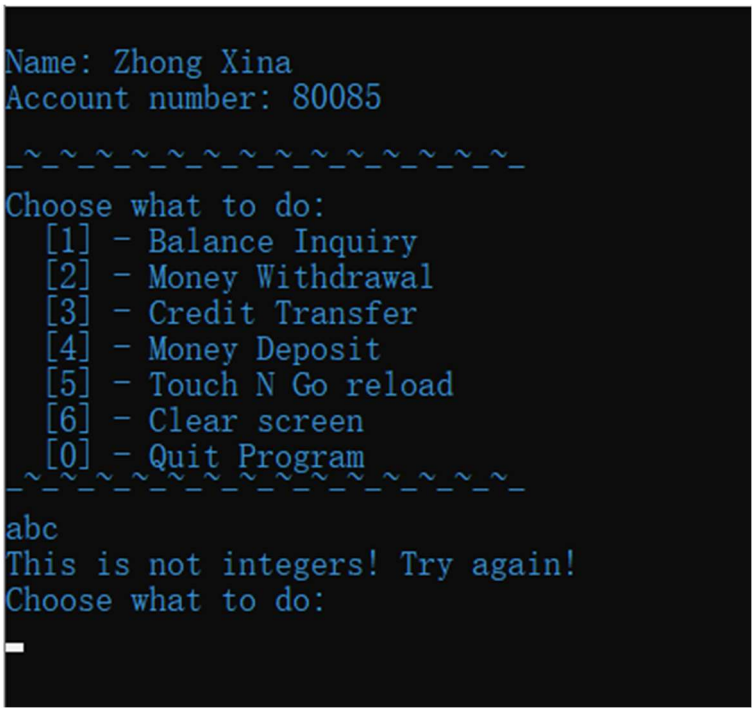


Testing

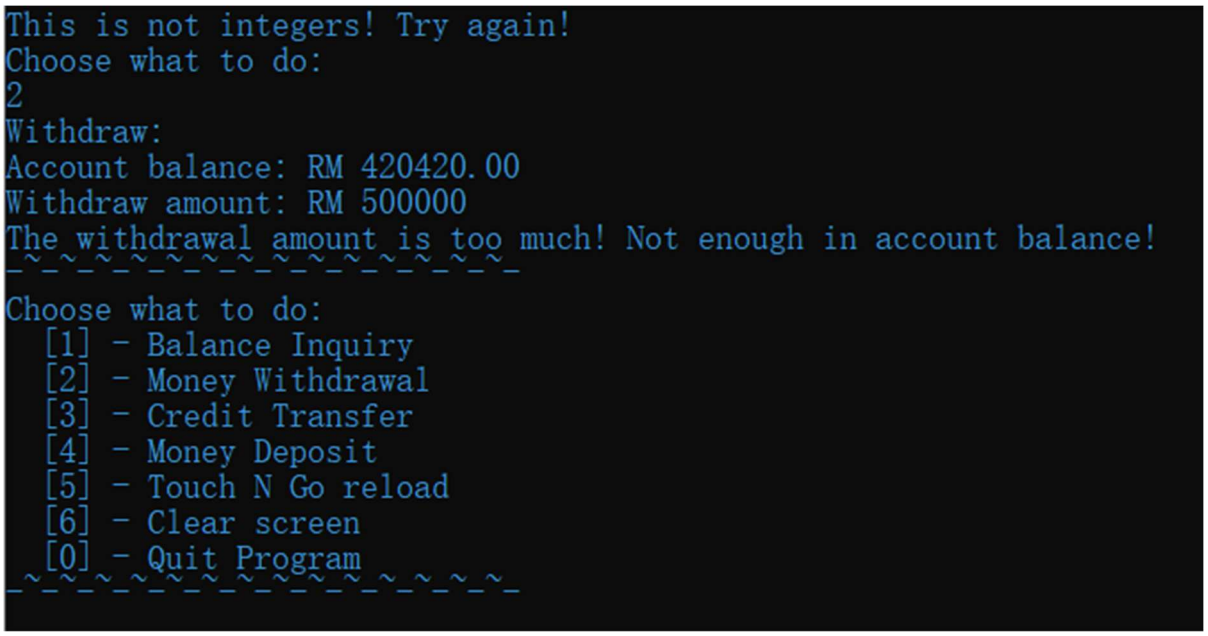
Wrong Pin:



Not integers (Character)



Overlimit



Account balance empty

